POINTS OF THE SPECTRUM OF A FIBRE PRODUCT OF RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- The Situation of a fibre product of rings, with B' = B x_A A' and the two structure maps. situation environment at source lines 521-542, immediately before the counted statement. Fixes all the rings used in the statement and in the proof.
- Spectrum of a ring and the induced maps between spectra, including the description of fibres over a prime. standard, used throughout the proof at lines 553-604. Supplies the points whose description is the statement.
- Prime ideals of a fibre product of rings and the correspondence with compatible pairs of primes. used in the two case discussions of the proof, lines 565-604. Provides the construction of the required primes and the verification of primeness.

Source reference. Original lemma 0B7J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14107–14107. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 08KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10885–10885. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0B1W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13909–13909. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01Z8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2583–2583. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In Situation 08KH we have*

$$\mathrm{Spec}(B') = \mathrm{Spec}(B) \amalg_{\mathrm{Spec}(A)} \mathrm{Spec}(A')$$

as topological spaces.

Proof. Since $B' = B \times_A A'$ we obtain a commutative square of spectra, which induces a continuous map

$$can : \mathrm{Spec}(B) \amalg_{\mathrm{Spec}(A)} \mathrm{Spec}(A') \longrightarrow \mathrm{Spec}(B')$$

as the source is a pushout in the category of topological spaces (which exists by Topology, Section 0B1W).

To show the map *can* is surjective, let $\mathfrak{q}' \subset B'$ be a prime ideal. If $I \subset \mathfrak{q}'$ (here and below we take the liberty of considering I as an ideal of B' as well as an ideal of A'), then $\mathfrak{q}'$ corresponds to a prime ideal of B and is in the image. If not, then pick $h \in I$, $h \notin \mathfrak{q}'$. In this case $B_h = A_h = 0$ and the ring map $B'_h \to A'_h$ is an isomorphism, see Lemma 01Z8. Thus we see that $\mathfrak{q}'$ corresponds to a unique prime ideal $\mathfrak{p}' \subset A'$ which does not contain I.

Since $B' \to B$ is surjective, we see that *can* is injective on the summand $\mathrm{Spec}(B)$. We have seen above that $\mathrm{Spec}(A') \to \mathrm{Spec}(B')$ is injective on the complement of $V(I) \subset \mathrm{Spec}(A')$. Since $V(I) \subset \mathrm{Spec}(A')$ is exactly the image of $\mathrm{Spec}(A) \to \mathrm{Spec}(A')$ a trivial set theoretic argument shows that *can* is injective.

To finish the proof we have to show that *can* is open. To do this, observe that an open of the pushout is of the form $V \amalg U'$ where $V \subset \mathrm{Spec}(B)$ and $U' \subset \mathrm{Spec}(A')$ are opens whose inverse images in $\mathrm{Spec}(A)$ agree. Let $v \in V$. We can find a $g \in B$ such that $v \in D(g) \subset V$. Let $f \in A$ be the image. Pick $f' \in A'$ mapping to f. Then $D(f') \cap U' \cap V(I) = D(f') \cap V(I)$. Hence $V(I) \cap D(f')$ and $D(f') \cap (U')^c$ are disjoint closed subsets of $D(f') = \mathrm{Spec}(A'_{f'})$. Write $(U')^c = V(J)$ for some ideal $J \subset A'$. Since $A'_{f'} \to A'_{f'}/IA'_{f'} \times A'_{f'}/JA'_{f'}$ is surjective by the disjointness just shown, we can find an $a'' \in A'_{f'}$ mapping to 1 in $A'_{f'}/IA'_{f'}$ and mapping to zero in $A'_{f'}/JA'_{f'}$. Clearing denominators, we find an element $a' \in J$ mapping to f^n in A. Then $D(a'f') \subset U'$. Let $h' = (g^{n+1}, a'f') \in B'$. Since $B'_{h'} = B_{g^{n+1}} \times_{A_{f^{n+1}}} A'_{a'f'}$ by a previously cited lemma, we see that $D(h')$ pulls back to an open neighbourhood of v in the pushout, i.e., the image of $V \amalg U'$ contains an open neighbourhood of the image of v. We omit the (easier) proof that the same thing is true for $u' \in U'$ with $u' \notin V(I)$. □